USAI - Nagpur City Command Center

AI-Powered Urban Safety Intelligence System

1. Executive Summary & Vision

USAI (Urban Safety AI) is an advanced spatial safety intelligence platform and 3D Command Center engineered for Nagpur City, India. Built for municipal command centers, police dispatchers (112), and emergency services, USAI integrates real-time risk heatmapping, CCTV blind spot detection, AI police force deployment, public event forecasting, and Explainable AI (XAI) into a unified digital twin.

2. System Architecture & Tech Stack

- Core Framework: Next.js 16 (App Router) + React 19 + TypeScript
- Styling: Tailwind CSS v4 + Cyan/Emerald Futuristic Theme
- Mapping & 3D Twin: Google Maps JS API + Mapbox GL JS 3D Extrusions
- GPU Acceleration: Deck.gl Heatmaps & Sensor Nodes
- Audio System: Web Audio API Alarm Synthesizer

3. Nagpur Administrative Sectors

- Sitabuldi (Z-SIT): 21.1458° N, 79.0882° E | Commercial Transit Hub | Pop: 120k
- Dharampeth (Z-DHA): 21.1412° N, 79.0610° E | Residential & Commercial | Pop: 180k
- Sadar (Z-SAD): 21.1610° N, 79.0815° E | Market & Dining District | Pop: 140k
- Itwari (Z-ITW): 21.1520° N, 79.1120° E | Dense Historic Commerce | Pop: 210k
- Sakkardara (Z-SAK): 21.1210° N, 79.1050° E | South Nagpur Residential | Pop: 195k
- Automotive Sq (Z-AUT): 21.1920° N, 79.0910° E | Highway & Industrial Corridor | Pop: 110k

4. Core Mathematics & Algorithmic Engines

Risk Score Formula: RiskScore = min(100, Round((IncidentLoad * 0.38 + TimeFactor * 0.27 + CCTVDeficit * 0.22 + LightingDeficit * 0.13) * 100))

Required Patrol Formula: RequiredPatrols = max(2, Round((Population / 50000) + (RiskScore / 15) + (CrimeVelocity / 20)))

5. REST API Architecture

- GET /api/incidents - Query incidents with zone, severity, and status filters.
- GET /api/cctv - Fetch surveillance camera health and operational coverage ratios.
- GET /api/zones - Spatial zone coordinates, officer details, and active patrol counts.
- GET /api/news - Real-time aggregated crime and safety news feed for Nagpur.

6. Quick Start & Execution

1. Run: npm install
2. Configure .env.local with NEXT_PUBLIC_MAPBOX_ACCESS_TOKEN
3. Run dev server: npm run dev